Adversarial verification of the quartic Maxwell theorem

Theorems 1.1–1.3, finite spheres, full-vector reproduction, and direct prior art

Independent AI-assisted audit prepared for Junjie Zheng

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Result. No defined counterexample was found within the measurable, bounded, reciprocal, nonmagnetic dielectric class with a strictly positive tensor loss floor. The upper inequality, finite-sphere sign enclosure, explicit sixth-order remainder, and every-resource-class lower construction were independently rederived. **One hypothesis is missing literally: susceptibility measurability.** The literature comparison needs substantive correction: fixed-smooth-shape quartic upper order and the two-dipole mechanism follow from earlier published formulas. This audit does not certify scientific priority.

The target is the released `OQ7_quartic_Maxwell_2026-10-10.tex`, SHA-256 beginning `0f4f6167725b10c8`. Its bytes match the source in the released archive. Neither original was edited. Three fresh reviewers began without previous audit verdicts; the parent independently checked calculations, wrote a second Maxwell solver, and inspected the cited primary equations. This is an AI mathematical and computational audit, not a proof-assistant certificate or independent human peer review.

Target	Audit disposition
Theorem 1.1	Proof and constants verified for measurable χ. The algebraic class alone omits that necessary hypothesis. Geometry uniformity survives, including disconnected and irregular supports.
Theorem 1.2	Nonzero coefficient interval and $21t^6/2000$ remainder reproduced for actual finite spheres. No dipole truncation is a premise of the proof.
Theorem 1.3	Homogeneous unequal-sphere positivity verified, including $M = m$. The supremum statement inherits the measurability qualification.
Novelty	Broad effect, construction mechanism, and fixed-smooth upper order have close antecedents. Priority for the precise uniform theorem remains unresolved.

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1 Findings that change the assessment

1.1 A literal missing hypothesis

The target calls D measurable and imposes, almost everywhere on D,

$$\chi^T = \chi, \quad \|\chi\|_{\mathrm{op}} \le M, \quad \mathrm{Im}_H\, \chi \succeq mI, \quad 0 < m \le M < \infty. \tag{1}$$

It never explicitly requires χ to be measurable or to belong to $L^\infty(D;\mathbb{C}^{3\times3})$. Its later statement that the methods need measurability and the caps does not specify the coefficient hypothesis. Both χ and χ^{-1} are used as L^2 multipliers; algebraic bounds do not imply measurability.

For a positive-volume closed ball D and a non-Lebesgue-measurable $N \subset D$, put $\chi = \mathrm{i}(1 + 1_N)I$ on D and zero outside. Reciprocity, $M = 2$, and $m = 1$ hold pointwise, but multiplying a nonzero constant vector by χ gives a nonmeasurable field. The constitutive operator and cross sections are undefined. This is a statement/definition gap, not a measurable medium violating the bound. The necessary hypothesis is explicitly $\chi \in L^\infty(D;\mathbb{C}^{3\times3})$, with (1) almost everywhere. All proofs below use that interpretation; the target was not silently amended.

The outgoing condition must apply to the *scattered* field. A total field containing a plane wave is not outgoing. The manuscript's construction makes this clear, so its shorthand is not an additional proof failure.

1.2 The prior-art comparison is incomplete

The target already disclaims certified priority, but does not compare the improvement directly with the relevant old Maxwell expansions. The 2014 Ammari–Deng–Millien formula implies quartic upper order for a fixed smooth lossy nonmagnetic shape. The 2015 Ammari–Ruiz–Yu–Zhang report gives an angular-amplitude version. Section 8 derives this inference rather than attributing to those papers a quartic theorem they do not explicitly state.

Hopkins et al. also contains an opposite-*linear*-polarization example and transverse two-dipole equations. Their static reduction gives the same positive homogeneous unequal-dipole interaction term as the manuscript. A distinction based mainly on complex polarization misses this close precedent. The exact uniform inequality and quantitative continuum certificates remain potentially distinct; not locating an identical theorem is not a priority certificate.

2 Independent geometry-uniform Maxwell derivation

Let $D \subset B_R(0)$ be measurable, $V = |D| > 0$, and χ measurable as above. Use vacuum outside, relative permeability I, and time convention $e^{-\mathrm{i}\omega t}$. For real unit e, d with $e \cdot d = 0$,

$$f_\pm(x) = e e^{\pm \mathrm{i} k d\cdot x}, \qquad Z_0 H_{\mathrm{inc},\pm} = \pm(d \times e) e^{\pm \mathrm{i} k d\cdot x}.$$

Both experiments use the same coefficient. On $\mathcal{H} = L^2(D;\mathbb{C}^3)$ let $\langle u, v\rangle = \int_D \bar{u}\cdot v$ and $[u, v] = \int_D u \cdot v$. Extend $p = \chi E$ by zero and set

$$\mathcal{K}_k p = (k^2 I + \nabla\nabla\cdot)(g_k * p)|_D, \qquad g_k(x) = \frac{e^{\mathrm{i}k|x|}}{4\pi|x|}, \tag{2}$$

$$\mathcal{L}_k = \chi^{-1} - \mathcal{K}_k, \qquad \mathcal{T}_k = \mathcal{L}_k^{-1}. \tag{3}$$

All derivatives are distributional.

2.1 The contact term and the dyadic estimate

The whole-space static multiplier is $-\xi\xi^T/|\xi|^2$, a negative orthogonal projection. Compression to D gives $\|\mathcal{K}_0\| \le 1$ and self-adjointness. The distributional Hessian includes

$$\partial_i\partial_j \frac{1}{4\pi r} = \mathrm{pv}\,\frac{3x_i x_j - r^2\delta_{ij}}{4\pi r^5} - \frac{\delta_{ij}}{3}\delta_0.$$

The contact terms cancel in $\mathcal{K}_k - \mathcal{K}_0$. For $u = kr$, put $b(u) = e^{\mathrm{i}u}(\mathrm{i}u - 1) + 1$. The radial and tangential kernel eigenvalues are

$$\frac{-2b(u)}{4\pi r^3}, \qquad \frac{u^2 e^{\mathrm{i}u} + b(u)}{4\pi r^3}.$$

Since $b'(u) = -ue^{\mathrm{i}u}$, $|b(u)| \le u^2/2$ for every $u \ge 0$. The kernel norm is at most $3k^2/(8\pi r)$. For $a_v = (3V/(4\pi))^{1/3}$, layer cake with $|D \cap B_s(x)| \le \min\{V, 4\pi s^3/3\}$ gives $\sup_x \int_D |x-y|^{-1}\,\mathrm{d}y \le 2\pi a_v^2$. Schur's test yields

$$\|\mathcal{K}_k - \mathcal{K}_0\| \le \frac{3}{4}k^2 a_v^2, \qquad \|\mathcal{L}_k\| \le L_k := m^{-1} + 1 + \frac{3}{4}k^2 a_v^2. \tag{4}$$

There is no hidden gap, aspect-ratio, component-count, or boundary constant.

2.2 All-angle radiation and coercivity

Define $\Pi_n = I - nn^T$ and $\mathcal{F}_k p(n) = \Pi_n \int_D p(x) e^{-\mathrm{i}kn\cdot x}\,\mathrm{d}x$. The angular representation of $\Im g_k$ proves exactly

$$\mathrm{Im}_H\,\mathcal{K}_k = \frac{k^3}{16\pi^2}\mathcal{F}_k^\dagger\mathcal{F}_k \succeq 0, \qquad \|\mathcal{F}_k\|^2 \le \frac{8\pi}{3}V. \tag{5}$$

For the norm bound apply Cauchy–Schwarz at fixed n and integrate $\int_{S^2}\Pi_n\,\mathrm{d}\Omega = 8\pi I/3$. Compactly supported L^2 currents are L^1, so Fubini applies to the smooth angular kernel. No approximation by smooth functions supported *inside* a possibly empty-interior D is needed.

The loss floor implies $\|\chi^{-1}\| \le 1/m$. Put

$$\rho = M^2/m, \qquad W = -\,\mathrm{Im}_H\,\chi^{-1} = \chi^{-\dagger}(\mathrm{Im}_H\,\chi)\chi^{-1} \succeq \rho^{-1} I. \tag{6}$$

Then

$$-\Im\langle p, \mathcal{L}_k p\rangle = \langle p, (W + \mathrm{Im}_H\,\mathcal{K}_k)p\rangle \ge \rho^{-1}\|p\|^2. \tag{7}$$

The bounded form $\langle u, \mathrm{i}\mathcal{L}_k v\rangle$ is complex coercive. Lax–Milgram gives surjectivity and injectivity, hence $\|\mathcal{T}_k\| \le \rho$. Equivalently, coercivity gives closed range, and the adjoint's same positive real part gives dense range. This verifies the inverse rather than just uniqueness. Negative real contrast and reciprocal nonnormal tensors do not defeat the estimate.

2.3 Equivalence to weak Maxwell scattering on arbitrary measurable sets

For $p = \mathcal{T}_k(f|_D)$ let $A_p = g_k * p$. Then $(\Delta + k^2)A_p = -p$ and $A_p \in H^2_{\mathrm{loc}}$ by local constant-coefficient elliptic regularity. The fields

$$E = f + (k^2 I + \nabla\nabla\cdot)A_p, \qquad U = Z_0 H = (\mathrm{i}k)^{-1}\nabla \times E$$

lie in $H_{\mathrm{loc}}(\mathrm{curl})$ and satisfy

$$\nabla \times E = \mathrm{i}kU, \qquad \nabla \times U = -\mathrm{i}k(E + p), \qquad p = 1_D \chi E$$

distributionally, with outgoing scattered fields. Conversely, subtracting this potential from any locally finite-energy scattering solution leaves an entire outgoing vacuum field, which vanishes by vacuum uniqueness on a containing smooth sphere. No normal, trace, finite perimeter, openness, or Lipschitz condition on D enters. Rough polarization divergence is allowed.

2.4 Joint sources, reciprocity, and the denominator

Unit-amplitude cross sections are exactly

$$A = k\langle p, Wp\rangle, \qquad S = \frac{k^4}{16\pi^2}\|\mathcal{F}_k p\|^2, \qquad X = k\Im\langle f, p\rangle = A + S. \tag{8}$$

These follow by dividing dissipated and radiated powers by incident intensity $1/(2Z_0)$ and using $E_\infty = k^2\mathcal{F}_k p/(4\pi)$. Transpose symmetry gives $[u, \mathcal{T}_k v] = [v, \mathcal{T}_k u]$. Since $f_- = \bar{f}_+$,

$$\langle f_+, \mathcal{T}_k f_+\rangle = [f_-, \mathcal{T}_k f_+] = [f_+, \mathcal{T}_k f_-] = \langle f_-, \mathcal{T}_k f_-\rangle.$$

Thus $X_+ = X_- = X$. The material is not conjugated. For two-column $F = (f_+, f_-)$ and $P = (p_+, p_-)$ the same equation gives $\mathrm{Im}_H(F^\dagger P) = P^\dagger(W + \mathrm{Im}_H\,\mathcal{K}_k)P$. Because $\|f_+\| = \sqrt{V} > 0$,

$$X \geq \frac{k}{\rho}\|p_+\|^2 \geq \frac{kV}{\rho L_k^2} > 0. \tag{9}$$

This derived bound vanishes with volume and imposes no fixed extinction threshold. Define $\Gamma = |A_+ - A_-|/X = |S_+ - S_-|/X$.

2.5 Assembling the quartic power and its constant

The common inverse gives $\|p_+ - p_-\| \leq 2\rho k R\sqrt{V}$. Set $h = (\rho X/k)^{1/2}$. Then $\|p_\pm\| \leq h$ and $\sqrt{V} \leq L_k h$. Difference of squared radiation norms gives

$$\begin{aligned} |S_+ - S_-| &\leq \frac{k^4 V}{6\pi}\|p_+ - p_-\|(\|p_+\| + \|p_-\|) \\ &\leq \frac{k^4 V}{6\pi}(2\rho k R\sqrt{V})(2h) \leq \frac{2}{3\pi}\rho^2 L_k k^4 R V X. \end{aligned} \tag{10}$$

Dividing by positive X and using $0 \leq A_\pm \leq X$ proves

$$\Gamma \leq \min\left\{1, \frac{2}{3\pi}\rho^2 L_k k^4 R V\right\}, \tag{11}$$

$$\Gamma \leq \boxed{\min\left\{1, \frac{8}{9}\rho^2\left(m^{-1} + 1 + \frac{3}{4}t^2\right)t^4\right\}}, \qquad t = kR. \tag{12}$$

The extra power comes from one bounded inverse acting on the incident difference. Every step holds at every $k, R > 0$. Material patterns may vary with R; no differentiability or convergence is used. Volume fractions may vanish arbitrarily fast.

3 The full-vector coefficient and the missing fifth power

Fix a reference support Ω and coefficient pattern, then shrink $D_R = R\Omega$ at fixed operating frequency. In reference coordinates use $\mathcal{T}_t, \mathcal{K}_t$ and set

$$q_0 = \mathcal{T}_0 e, \quad q_1 = \mathcal{T}_0[(d\cdot y)e], \quad p_j = \int_\Omega q_j, \quad D_0 = \Im(e\cdot p_0) = \langle q_0, W q_0\rangle > 0.$$

The affine input need not be curl free. It is a legitimate vector L^2 input to the static volume inverse; replacing it by a scalar-potential ansatz would need proof.

Let $c = e\cos(td\cdot y)$, $s = e\sin(td\cdot y)$, $u = \mathcal{T}_t c$, and $v = \mathcal{T}_t s$. The exact currents are $q_\pm = u \pm \mathrm{i}v$. The resolvent identity gives

$$\mathcal{T}_t - \mathcal{T}_0 = \mathcal{T}_t(\mathcal{K}_t - \mathcal{K}_0)\mathcal{T}_0 = O(t^2), \qquad u = q_0 + O(t^2), \quad v = tq_1 + O(t^3)$$

in operator/vector L^2 norms. These constants concern the fixed pattern; the universal upper bound was proved separately without fixing a pattern. The exact radiation operator has kernel

$$Q_t(x,y) = \frac{1}{16\pi^2}\int_{S^2} \Pi_n e^{\mathrm{i}tn\cdot(x-y)}\,\mathrm{d}\Omega_n. \tag{13}$$

Substitution $n \mapsto -n$ makes it real and even. Hence $Q_t - Q_0 = O(t^2)$ and $\langle u, Q_0 v\rangle = (\int u)^\dagger(\int v)/(6\pi)$. Exact change of physical units gives

$$S_\pm = k^4 R^6 \langle q_\pm, Q_t q_\pm\rangle, \qquad X = kR^3\Im\{[c,\mathcal{T}_t c] + [s,\mathcal{T}_t s]\}.$$

Reciprocity cancels the mixed forward terms. Therefore

$$\frac{S_+ - S_-}{k^4R^6} = -4\Im\langle u, Q_t v\rangle = -\frac{2}{3\pi}t\Im(p_0^\dagger p_1) + O(t^3), \tag{14}$$

$$\frac{X}{kR^3} = D_0 + O(t^2). \tag{15}$$

It follows that

$$\boxed{\Gamma(t) = Ct^4 + O(t^6), \qquad C = \frac{2|\Im(p_0^\dagger p_1)|}{3\pi D_0}.} \tag{16}$$

No t^5 error is hidden: the signed radiation form has $O(t^3)$ error before multiplication by t^3, and the denominator changes by $O(t^2)$. The inequality $||a|-|b|| \le |a-b|$ also covers a vanishing leading coefficient. Observation phase cancellation occurs after exact full-angle integration, not pointwise. Magnetic dipoles, electric quadrupoles, and all higher currents remain in $\mathcal{T}_t$ and Q_t.

4 Actual heterogeneous spheres: noncancellation

Take reference centers $\mp e_z/2$, radii $a, 2a$ with $a = 1/20$, and $\chi_1 = \mathrm{i}I$, $\chi_2 = 2\mathrm{i}I$. Let $v_1 = 4\pi a^3/3$, $v_2 = 8v_1$, $w = 12\pi a^3$ and $b = 3/5$. The surface gap is $17/20$ and $|y| \le b$ on the support.

4.1 Exact isolated-sphere response for arbitrary input

On a ball, $\mathcal{K}_0 e = -e/3$. If $A_j = (\chi_j^{-1} - \mathcal{K}_{0,jj})^{-1}$, then $A_j e = \beta_j e$, where $\beta_j = 3\chi_j/(3+\chi_j)$. Bilinear symmetry, tested against each constant real vector, yields for every $h \in L^2$,

$$\int_{\Omega_j} A_j h = \beta_j \int_{\Omega_j} h. \tag{17}$$

There is no conjugate on β_j. An independent vector solution checks the affine input. In centered coordinates put

$$h_s = \tfrac{1}{2}(\xi_z e_x + \xi_x e_z), \qquad h_a = \tfrac{1}{2}(\xi_z e_x - \xi_x e_z).$$

The Newton potential of $\xi_j 1_{B_s}$ is $(s^2/6 - |\xi|^2/10)\xi_j$ inside and $s^5\xi_j/(15|\xi|^3)$ outside. Differentiating gives

$$\mathcal{K}_0 h_s = -\tfrac{2}{5}h_s, \quad \mathcal{K}_0 h_a = 0, \qquad A_j(\xi_z e_x) = \frac{5\chi_j}{5+2\chi_j}h_s + \chi_j h_a.$$

The antisymmetric part carries the curl and was retained. Both parts have zero moment; the antisymmetric part has zero exterior static field.

4.2 Enclosing all intersphere interactions

Between the spheres the static kernel is

$$H(r) = \frac{3rr^T - |r|^2 I}{4\pi|r|^5}, \qquad \|H(r)\| = \frac{1}{2\pi|r|^3}.$$

With $A_b = \operatorname{diag}(A_1, A_2)$, the off-diagonal block operator has norm at most

$$c_o = \frac{\sqrt{v_1 v_2}}{2\pi(17/20)^3} = \frac{4\sqrt{2}}{14739} < \frac{1}{2500}.$$

The block norm is the maximum of the two cross-block norms, not their sum. Since $\|A_b\| \le 2$,

$$\delta := \|\mathcal{T}_0 - A_b\| \le \frac{4c_o}{1-2c_o} < \frac{1}{600}. \tag{18}$$

This is an estimate on the entire current space, including every static multipole and repeated interaction. It is not a dipole-ansatz error.

The isolated moments (all parallel to e_x) are

$$\alpha_1 = 4\pi a^3 \frac{1+3\mathrm{i}}{10}, \qquad \alpha_2 = 32\pi a^3 \frac{4+6\mathrm{i}}{13}, \quad p_0^b = \alpha_1 + \alpha_2, \quad p_1^b = (\alpha_2 - \alpha_1)/2.$$

Exact rational arithmetic gives

$$I_b := \Im((p_0^b)^\dagger p_1^b) = -\frac{8}{195}w^2, \qquad D_b = \frac{173}{130}w.$$

Integration has norm $\sqrt{w}$. Bounds on both moments, including the product of their errors, give

$$|I_0 - I_b| \le b(4\delta + \delta^2)w^2, \qquad |D_0 - D_b| \le \delta w, \qquad I_0 = \Im(p_0^\dagger p_1).$$

Substituting $\delta = 1/600$ into the bound gives the relative-error majorant $31213/320000 = 0.097540625 < 1/10$. Hence $I_0 < 0$ for the actual finite spheres, and

$$\frac{12}{325}w^2 \le |I_0| \le \frac{44}{975}w^2, \qquad \frac{10367}{7800}w \le D_0 \le \frac{10393}{7800}w. \tag{19}$$

Substituting in (16) proves

$$\boxed{\frac{36}{1299125} \le C \le \frac{44}{1295875}}, \qquad 2.771095 \times 10^{-5} \le C \le 3.395390 \times 10^{-5}. \tag{20}$$

The signed scattering difference is positive for sufficiently small t with this ordering; the absorption difference is negative. This enclosure does not identify the exact value or an optimal class-wide prefactor.

5 Independent explicit sixth-order remainder

Use the conservative diameter $L = 6/5$. For $t \le 1$, the dynamic-minus-static kernel has numerator eigenvalues

$$A(s) = e^{\mathrm{i}s}(s^2 + \mathrm{i}s - 1) + 1, \qquad B(s) = 2\{e^{\mathrm{i}s}(1 - \mathrm{i}s) - 1\}, \qquad s = t|x - y|.$$

Since $A'(s) = s(1 + \mathrm{i}s)e^{\mathrm{i}s}$, $B'(s) = 2se^{\mathrm{i}s}$, and $s \le 6/5 < \sqrt{3}$, both numerator magnitudes are at most s^2. The kernel norm is bounded by $t^2/(4\pi|x - y|)$. Schur integration in a ball of radius L and the actual $W \succeq I/2$ imply

$$\|\mathcal{K}_t - \mathcal{K}_0\| \le \frac{18}{25}t^2, \qquad \|\mathcal{T}_t - \mathcal{T}_0\| \le \frac{72}{25}t^2, \qquad \|\mathcal{T}_t\| \le 2. \tag{21}$$

The last estimate retains exact outgoing radiation. Taylor inequalities then give

$$\|u - q_0\| \le \frac{81}{25}t^2\sqrt{w}, \qquad \|v - tq_1\| \le \frac{9}{5}t^3\sqrt{w}. \tag{22}$$

Also $\|u\| \le 2\sqrt{w}$, $\|v\| \le 2bt\sqrt{w}$, $\|q_0\| \le 2\sqrt{w}$, $\|q_1\| \le 2b\sqrt{w}$, and

$$\|Q_0\| \le \frac{w}{6\pi}, \qquad \|Q_t - Q_0\| \le \frac{wL^2}{8\pi}t^2.$$

For the latter bound use $1 - \cos s \le s^2/2$ in (13) and Schur's test. Decompose

$$\langle u, Q_t v\rangle - t\langle q_0, Q_0 q_1\rangle = \langle u, (Q_t - Q_0)v\rangle + \langle u - q_0, Q_0 v\rangle + \langle q_0, Q_0(v - tq_1)\rangle.$$

The three bounds, in units w^2t^3/π, are $54/125$, $81/125$, and $3/5$. Their sum is $42/25$. For $\mathscr{S} = (S_+ - S_-)/(k^4R^6)$ and $\mathscr{S}_0 = -2I_0/(3\pi)$,

$$|\mathscr{S} - t\mathscr{S}_0| \le \frac{168}{25\pi}w^2t^3. \tag{23}$$

The extinction constant includes both cosine and sine terms. With $\mathscr{D} = X/(kR^3)$,

$$\begin{aligned} |[c, u] - [e, q_0]| &\le \|c - e\|\|u\| + \|e\|\|u - q_0\| \le (b^2 + 81/25)wt^2, \\ |[s, v]| &\le 2b^2wt^2. \end{aligned}$$

Therefore

$$|\mathscr{D} - D_0| \le \frac{108}{25}wt^2. \tag{24}$$

For $t \le 1/4$, (19) and (24) give both $D_0 \ge w$ and $\mathscr{D} \ge w$. Since $|\mathscr{S}_0| \le 88w^2/(2925\pi)$, division gives

$$\begin{aligned} |\Gamma - Ct^4| &\le \frac{w}{\pi}\left\{\frac{168}{25} + \frac{88}{2925}\frac{108}{25}\right\}t^6 \\ &= \frac{w}{\pi}\frac{55656}{8125}t^6 < \frac{7w}{\pi}t^6 = \boxed{\frac{21}{2000}t^6}, \qquad 0 < t \le \frac{1}{4}. \end{aligned} \tag{25}$$

The independent checker reproduces every rational comparison, including $\frac{21}{2000}(1/40)^2 < \frac{1}{4}\frac{36}{1299125}$. Hence $\Gamma(t) \ge 27t^4/1299125$ for $0 < t \le 1/40$. This is an analytic full-Maxwell remainder, independently of Mie numerics.

Radius convention. Here R is center separation and a valid enclosing radius. The minimum enclosing radius is $R_{\min} = 23R/40$, centered at $Re_z/40$. For $\tau = kR_{\min}$, multiply the coefficient by $(40/23)^4$ and the remainder coefficient by $(40/23)^6$. The intervals become $\tau \le 23/160$ for the remainder and $\tau \le 23/1600$ for the lower bound. The target's coefficient conversion is correct; converting the entire inequality requires all these changes.

6 Every resource class: homogeneous spheres

Fix $\mu > 0$ and put $\chi = \mathrm{i}\mu I$ on both balls, still centered at $\mp e_z/2$ with radii $a, 2a$, now choosing one sufficiently small fixed $a \le 1/20$. Set $v_1 = 4\pi a^3/3$, $v_2 = 8v_1$, $w = 12\pi a^3$, $h = 1 - 3a \ge 4/5$, and $\beta = 3\mathrm{i}\mu/(3 + \mathrm{i}\mu)$. The isolated and full static inverses A, T satisfy $\|A\|, \|T\| \le \mu$, and the off-diagonal operator satisfies $\|C_o\| \le 4a^3$. Iterating the exact resolvent identity twice gives

$$T = A + AC_oA + AC_oAC_oT, \qquad \|T - A - AC_oA\| \le \mu^3\|C_o\|^2. \tag{26}$$

This does not presume Neumann-series convergence.

6.1 The leading interaction and affine source

Harmonic mean value in each variable of $H(x - y)$ gives exactly $\int_{\Omega_i}\int_{\Omega_j} H(x - y) = v_iv_jH(c_i - c_j)$ for disjoint balls. Put $g_0 = -1/(4\pi)$ and write moments through one interaction as $\widetilde{p}_j = p_j^b + p_j^{(1)}$. Then

$$p_0^b = \beta w e_x, \quad p_1^b = \tfrac{1}{2}\beta(v_2 - v_1)e_x, \quad p_0^{(1)} = 2g_0\beta^2 v_1v_2e_x, \quad |\widetilde{p}_0| \le 2|\beta|w.$$

The isolated product is real. The first constant-drive correction gives

$$\mathscr{L} := \Im((p_0^{(1)})^\dagger p_1^b) = -g_0v_1v_2(v_2 - v_1)|\beta|^2\Im\beta = \frac{2688\pi^2\mu^3}{(9 + \mu^2)^2}a^9 > 0. \tag{27}$$

The center terms of $p_1^{(1)}$ cancel, but its complete affine contribution does not. On sphere j let $q_j^o = A_j[(z - z_j)e_x]$. Then

$$\int q_j^o = 0, \quad \|q_j^o\| \le \mu r_j\sqrt{v_j/5}, \quad \int |y - c_j||q_j^o(y)|\,\mathrm{d}y \le \frac{\mu\sqrt{3}}{5}r_j^2v_j.$$

Direct differentiation gives $\|DH(r)[u]\| \le 6/(\pi|r|^4)$ for a real unit u. Subtract $H(c_i - c_j)$ using the zero moment. The relevant segments remain at distance at least h. Receiving-ball integration by (17) gives

$$|p_1^{(1)}| \le \frac{6\sqrt{3}|\beta|\mu}{\pi h^4}a^2v_1v_2.$$

It follows that

$$\frac{|\Im(\widetilde{p}_0^\dagger\widetilde{p}_1) - \mathscr{L}|}{\mathscr{L}} \le \frac{48\sqrt{3}}{7h^4}(9 + \mu^2)a^2 < 30(9 + \mu^2)a^2. \tag{28}$$

An additional audit calculation evaluates the exact first affine correction. With $\eta = 5\mathrm{i}\mu/(5 + 2\mathrm{i}\mu)$, the explicit affine sphere solution gives $p_1^{(1)} = (32\pi/5)\beta\eta a^8e_x$. Both $\Im((p_0^b)^\dagger p_1^{(1)})$ and $\Im((p_0^{(1)})^\dagger p_1^{(1)})$ are positive, since

$$\Im\eta = \frac{25\mu}{25 + 4\mu^2} > 0, \qquad \Im(\bar{\beta}\eta) = -\frac{15\mu^3}{(9 + \mu^2)(25 + 4\mu^2)} < 0.$$

They reinforce the leading sign. This is not a claim that all subsequent interactions have the same sign.

6.2 All remaining interactions and the order of limits

Let $\delta_2 = \mu^3\|C_o\|^2$. Equation (26) gives $|p_0 - \widetilde{p}_0| \le \delta_2 w$ and $|p_1 - \widetilde{p}_1| \le b\delta_2 w$, with $b = 3/5$. Using one *exact* factor in the difference of products includes all error products:

$$\begin{aligned}|\Im(p_0^\dagger p_1) - \Im(\widetilde{p}_0^\dagger \widetilde{p}_1)| &\le b\delta_2(\mu + 2|\beta|)w^2 \\ &\le \frac{9}{5}\mu^4\|C_o\|^2 w^2.\end{aligned}$$

Combining with (28) and $\|C_o\|^2 \le 16a^6$ yields

$$\frac{|\Im(p_0^\dagger p_1) - \mathscr{L}|}{\mathscr{L}} \le 30(9+\mu^2)a^2 + \frac{54}{35}\mu(9+\mu^2)^2a^3. \tag{29}$$

Choose, once before $t \to 0$,

$$0 < a \le \min\left\{\frac{1}{20}, [120(9+\mu^2)]^{-1/2}, \left[\frac{35}{216\mu(9+\mu^2)^2}\right]^{1/3}\right\}. \tag{30}$$

The relative error is at most $1/2$. Also $\mu w/(1+\mu)^2 \le D_0 = \mu^{-1}\|Te_x\|^2 \le \mu w$. The fixed-pattern coefficient theorem therefore gives

$$\Gamma = C_h t^4 + O(t^6), \qquad \frac{224\mu^2 a^6}{3(9+\mu^2)^2} \le C_h \le \frac{224\mu^2(1+\mu)^2 a^6}{(9+\mu^2)^2}. \tag{31}$$

Set $\mu = m$. This satisfies every $M \ge m$, including equality. The positive lower coefficient and (12) prove the claimed $\Theta_{m,M}(t^4)$ class supremum in the measurable-coefficient class. There is no simultaneous uncontrolled separation/size limit. Theorem 1.3 need not have the heterogeneous example's numerical remainder constant.

6.3 A serious attempted cancellation: static normality

For homogeneous $\chi = \mathrm{i}\mu$, the static Green operator is real self-adjoint. Thus $T_0^\dagger T_0$ is real, and purely static currents have equal unweighted norms under conjugate drives. This fact is true. It does not imply $\Im(p_0^\dagger p_1) = 0$, which uses $T_0^\dagger Q_0 T_0$: Q_0 need not commute with $\mathcal{K}_0$.

Retaining radiation resolves the physical absorption difference. Write $\mathcal{K}_t = H_t + \mathrm{i}t^3 Q_t$, with H_t real self-adjoint, and put $S_t = (-H_t - \mathrm{i}/\mu)^{-1}$. The reactive inverse is normal, whereas

$$T_t = S_t + \mathrm{i}t^3 S_t Q_t S_t + O(t^6)$$

can make $T_t^\dagger T_t$ nonreal at order t^3. The exact identity

$$\mathrm{Im}_H\, T_t = T_t^\dagger(\mu^{-1}I + t^3 Q_t)T_t$$

has real matrix elements between real c, s by transpose reciprocity. Thus $\mu^{-1}\Im\langle u, v\rangle + t^3\Im\langle u, Q_t v\rangle = 0$, and exactly

$$\frac{A_+ - A_-}{kR^3} = 4t^3\Im\langle u, Q_t v\rangle, \qquad \frac{S_+ - S_-}{kR^3} = -4t^3\Im\langle u, Q_t v\rangle.$$

These are respectively $\pm 2I_0 t^4/(3\pi) + O(t^6)$. Radiation supplies the first absorption difference. The radiative current correction $T_t - S_t$ changes the signed scattering difference only at $O(t^7)$ in these units; reactive corrections can enter at $O(t^6)$. The full follow-up derivation is included in the archive.

Material realizability. At a fixed operating frequency $\omega_0 > 0$, for any $\gamma > 0$ the scalar laws

$$\chi_j(\omega) = \frac{j\gamma\omega_0}{\omega_0^2 - \omega^2 - \mathrm{i}\gamma\omega}, \qquad j = 1, 2,$$

are passive causal Lorentz responses and equal $\mathrm{i}j$ at ω_0. Replacing j by μ realizes the homogeneous example. Their poles are below the real axis, the positive-frequency imaginary part is positive, and $z\chi(z)$ has positive imaginary part for $\Im z > 0$. Geometry shrinks while the material functions remain fixed. No loss floor at all frequencies down to zero is being asserted.

7 Full-vector numerical reproduction and independent controls

The unmodified released `reproduce.py` completed successfully in a separate extraction, including its 25-case sphere sweep, isolated-sphere Mie checks, algebra, and figures. The 25 recorded results reproduced identically. This checks replayability, not implementation independence.

The new `independent_numerics/multiparticle_mie.py` imports no released solver code. It eliminates sphere boundary conditions analytically and projects translated fields on complete angular harmonics, instead of the released boundary-matching solve. Multiparticle spherical-wave methods are established; Xu's sphere-aggregate formulation is a method antecedent, not a novelty claim for the new code [7].

7.1 The independently checked formulation

For orthonormal harmonics $\Psi = \nabla_S Y/\sqrt{\ell(\ell+1)}$ and $\Phi = n \times \Psi$, the code uses

$$M = -\mathrm{i} z_\ell \Phi, \qquad N = \mathrm{i}\left[\sqrt{\ell(\ell+1)}\frac{z_\ell(kr)}{kr} Y n + \left(z_\ell'(kr) + \frac{z_\ell(kr)}{kr}\right)\Psi\right].$$

They satisfy $\nabla \times M = kN$, $\nabla \times N = kM$. Tangential Maxwell boundary conditions give $T_M = -b_\ell$, $T_N = -a_\ell$ for the usual Mie coefficients, evaluated through Riccati–Bessel logarithmic derivatives. All $-\ell \le m \le \ell$, both families, and $1 \le \ell \le \ell_{\max}$ are retained. Outgoing fields are projected on a sphere within the receiving center's analytic neighborhood. Unknowns are balanced by $k^{\ell+2}/(2\ell-1)!!$ before solution.

Regular-wave flux splitting and the forward optical theorem independently give

$$X = -k^{-2}\Re \sum_i a_{\mathrm{inc},i}^\dagger b_i, \qquad A = k^{-2} \sum_{i,\ell,m,P} (-\Re T_{i\ell P} - |T_{i\ell P}|^2)|a_{\mathrm{exc},i\ell m P}|^2.$$

The scattering calculation adds full far fields with center phases $e^{-\mathrm{i}k n\cdot c_i}$ before angular integration. The review found no missing factor two, impedance, 4π, or interference term. The independent read-only formulation report is included separately.

The $\ell_{\max} = 8$ run at $t = 0.1$ agrees with $\ell_{\max} = 6$ to the printed precision. Over the heterogeneous sweep, the largest normalized energy residual was 8.6432×10^{-16} and reciprocity residual 6.9107×10^{-16}. The new standard-library algebra checker passed 86 checks. Exact arithmetic validates its scalar expressions; the preceding analysis supplies the operator proofs.

7.2 Controls targeting concrete failure modes

Changing translation probe fraction from 0.3 to 0.2 and 0.4 while increasing angular quadrature changed Γ relatively by less than 3×10^{-13} at $t = 0.1$. A generic rigid rotation of geometry, incidence, and

Table 1: Independent heterogeneous finite-sphere results, $R = 1$ and $\ell_{\max} = 6$. The first row is outside the explicit $t \le 1/4$ remainder interval.

$t = kR$	Γ/t^4	Relative difference from released Γ
0.400	$2.893456409632 \times 10^{-5}$	5.98×10^{-13}
0.200	$3.030480210012 \times 10^{-5}$	2.74×10^{-12}
0.100	$3.065512287396 \times 10^{-5}$	3.49×10^{-13}
0.050	$3.074318778290 \times 10^{-5}$	1.11×10^{-11}
0.025	$3.076523360320 \times 10^{-5}$	3.71×10^{-11}

polarization excited formerly absent azimuthal modes and changed Γ by less than 10^{-12} relatively. Transverse-incidence mirror symmetry returned zero asymmetry. Equal identical spheres and a lossless unequal pair returned asymmetry at roundoff. Three isolated-sphere cases agreed with separate analytic Mie references to within 2×10^{-14} relatively in both extinction and scattering.

For a homogeneous pair with radii $0.025, 0.05$ and $\chi = \mathrm{i}$, the signed scattering asymmetry is opposite to the heterogeneous pair. At $t = 0.2$, 0.1, 0.05, and 0.025, respectively, its values are

$$\Gamma/t^4 \simeq 10^{-10}(6.00585,\ 4.54880,\ 4.17993,\ 4.08742).$$

The analytic coefficient enclosure is $[1.8229167 \times 10^{-10}, 2.1875 \times 10^{-9}]$.

These are floating-point comparisons, not interval enclosures of the infinite series. The equation residual tests an already assembled finite matrix, not omitted multipoles, projection quadrature, special functions, or subtraction of powers. Tiny Bessel denominators, internal zeros of logarithmic derivatives, and cancellation in the magnetic Mie coefficient are known accuracy limitations of this implementation. At the smallest homogeneous point, $\Gamma \approx 1.60 \times 10^{-16}$ is below the energy residual normalized by extinction (6.90×10^{-16}); that point alone cannot validate the sign by global conservation residuals. The rigorous sign and remainder come from the analytic enclosure, not this weak numerical signal. No certified Mie truncation bound was produced.

8 Direct comparison with prior small-particle theory

8.1 Quartic upper order implicit in earlier Maxwell expansions

Ammari–Deng–Millien's first arXiv version is dated 11 December 2014. Its Theorem 5.8, Eq. (5.32), gives the leading electric and magnetic static polarization terms with an $O(\delta^4)$ field remainder for a fixed smooth scaled body [1]. The 2015 Ammari–Ruiz–Yu–Zhang institutional report repeats the expansion in Theorem 4.1 and supplies angular amplitudes in Proposition 4.1, with real transverse incident polarization in Theorem 4.2 [2].

The following is this audit's inference from those formulas. Fix frequency, a smooth homogeneous reference B, finite scalar contrast with positive imaginary part, and $D_\delta = \delta B$. At zero magnetic contrast the leading magnetic polarization tensor vanishes: the underlying layer system is regular in the equal-permeability limit. Both electric incident fields have value e at the shrinking center. Their full angular far fields therefore have the same leading term:

$$F_\pm(n) = \delta^3 F_0(n) + O(\delta^4), \qquad S_+ - S_- = O(\delta^7).$$

Static dissipation gives $D_0 = \int_B \Im\chi\, |E_0|^2 > 0$ and $X = k\delta^3(D_0 + O(\delta))$. Consequently

$$\Gamma = O(\delta^4) = O((k\delta)^4) \quad \text{at fixed } k.$$

This deduction does not require the older papers to use Γ or explicitly call the order quartic. It supplies a fixed-smooth-shape antecedent. It does not provide the constants or geometric quantifier of (12). The old resonance-parameter uniformity cannot be substituted for uniformity over all measurable supports. No term-by-term identification of every older vector response tensor with the manuscript's continuum coefficient was established here.

8.2 Old transverse two-dipole equations give the same mechanism

Hopkins et al., Eqs. (12)–(15), explicitly gives a retarded transverse two-electric- dipole system. Its Fig. 3, printed p. 753, uses opposite linearly polarized waves for same-material gold stacked crosses, reporting absorption, total scattering, and extinction [3]. Markel's Eqs. (8), (42), and (44)–(49) give consistent coupled-dipole energy relations [4].

Here is the audit's algebraic reduction in volume-polarizability units. For dimensionless static α_j, unit center separation, and transverse coupling $g = -1/(4\pi)$, let $\Delta = 1-g^2\alpha_1\alpha_2$. The uniform- and affine-drive total moments are

$$P_0 = \frac{\alpha_1 + \alpha_2 + 2g\alpha_1\alpha_2}{\Delta}, \qquad P_1 = \frac{\alpha_2 - \alpha_1}{2\Delta}. \tag{32}$$

Direct multiplication gives

$$I_{\mathrm{CDA}} = \Im(P_0^* P_1) = \frac{\Im(\alpha_1^*\alpha_2) + g(|\alpha_1|^2\Im\alpha_2 - |\alpha_2|^2\Im\alpha_1)}{|\Delta|^2}. \tag{33}$$

The retarded rescaled coupling has no linear term in t; its reactive correction starts at t^2 and radiation at t^3. The angular power is even in observation phase. The same cosine/sine algebra yields within this model

$$\Gamma = C_{\mathrm{CDA}} t^4 + O(t^6), \qquad C_{\mathrm{CDA}} = \frac{2|I_{\mathrm{CDA}}|}{3\pi\Im P_0}.$$

The full model must retain radiation reaction for energy consistency. Combining uncoupled absorption with coherently summed scattering is not a valid replacement.

For one material $\alpha_j = v_j\beta$, $v_2 > v_1$, the numerator becomes

$$I_{\mathrm{CDA}} = \frac{-g v_1 v_2 (v_2 - v_1)|\beta|^2 \Im\beta}{|1 - g^2 v_1 v_2 \beta^2|^2} > 0. \tag{34}$$

For $\chi = \mathrm{i}\mu$, $v_2 = 8v_1$, its leading small-a value is exactly $2688\pi^2\mu^3 a^9/(9+\mu^2)^2$, matching (27). The basic homogeneous noncancellation is thus already a consequence of the old point model. For the heterogeneous parameters, the scalar static model gives $C_{\mathrm{CDA}} = 63880/2075856051 \approx 3.077284669 \times 10^{-5}$. This is not the exact finite-sphere coefficient. Taking t small alone does not remove a fixed static multipole error when the radius/gap ratio is fixed; Sections 4–6 provide the continuum upgrade.

8.3 Unresolved priority and access limits

Sounas–Alù (2014) is a close antecedent for reciprocal extinction symmetry and lossy unequal-sphere asymmetry [5]. Its first page through Eq. (2), official abstract, and figure captions were inspected; the remaining pages and Eqs. (3)–(4) were not obtained. Ammari–Khelifi (2003) remains an abstract-only mathematical lead, and Mann–Alù (2023) an abstract/program lead. These are material access limits, not evidence of absent results. Brown et al. (2010) is earlier physical heterodimer work [6]; the inspected full article text uses finite collection and substrate modeling. It supports physical precedence but does not supply the present vacuum class theorem in the inspected material.

A targeted counterexample lead, Ylä-Oijala et al. (2025), was also checked [8]. Its accessible first page states that its extinction-symmetry-breaking examples use nonreciprocal materials; Fig. 1 specifies equal extinction for the opposite-wave-vector, conjugate-electric-field reciprocal pair. This includes the target's real polarization. The complete accepted manuscript was inaccessible, so no uninspected constitutive tensors are attributed to it.

Claim component	Direct comparison
Reciprocal equal extinction and opposite total absorption/scattering asymmetry	Established antecedents.
Fixed-smooth nonmagnetic quartic upper order	Follows from older full-Maxwell formulas under fixed positive loss.
Two-dipole quartic mechanism and homogeneous unequal-dipole sign	Derived above from the old equations.
All-measurable-support exact inequality under m, M, R, V	No matching theorem located; priority not certified.
Actual-sphere noncancellation and $21t^6/2000$ remainder	Additional continuum certification beyond a point model; priority unresolved.
Every-m, M supremum theorem	Verified with the completed material definition; no matching prior statement located.

The literature section needs an equation-level revision acknowledging the implicit fixed-smooth quartic consequence and Hopkins's opposite-linear example. The evidence does not support a newly discovered asymmetry effect, unequal-sphere mechanism, or first quartic consequence of Maxwell theory. Whether the precise uniform quantitative theorem is new remains unresolved.

9 Counterexample ledger and conclusion limits

The attempts below target the advertised class without changing its resources.

Failure mechanism	Audit result
Nonmeasurable coefficient obeying pointwise caps	Exposes a missing hypothesis: the multiplier and cross sections may be undefined. No physical bound violation is asserted.
Disconnected, empty-interior, infinitely many-component, or irregular support	Zero extension, Fourier projection, Schur bounds, and whole-space weak Maxwell construction need no boundary regularity.
Nearly touching components or vanishing volume fraction	No separation constant enters the upper theorem; the derived denominator estimate controls division as volume collapses.
Negative real contrast, resonance, nonnormal tensors	$W \succeq m/M^2$ bounds the exact inverse under the fixed caps.
Material patterns varying with size	No coefficient derivative, compactness, or fixed-pattern hypothesis enters the upper inequality.
Different material for opposite waves	Never used: both currents are images under one exact inverse.
Omitted magnetic or quadrupolar radiation	The exact radiation map and vector inverse contain all multipoles. Rotation of the numerical example adds an independent mode check.
Non-curl-free affine input	The explicit symmetric/antisymmetric sphere solution retains its curl and confirms the arbitrary-input moment identity.

Failure mechanism	Audit result
Hidden t^5 correction	Operator-norm $O(t^2)$ and the even radiation kernel give the $O(t^6)$ normalized remainder.
Static normality forces homogeneous cancellation	It constrains unweighted norms, not the radiated quadratic form. The radiation term supplies the absorption difference.
Point-model sign destroyed by multipoles	Full-current operator enclosures certify the sign for fixed finite spheres; decreasing t alone is not used to remove static errors.
$M = m$ prohibits heterogeneous choices	The single material $\chi = imI$ suffices with the fixed radius choice.
2025 extinction-symmetry-breaking examples	Accessible primary text uses nonreciprocal materials and retains equality for the target reciprocal source pair.

Magnetic constitutive response, bianisotropy, spatial dispersion, loss floors on only some components, general elliptical polarization, and material caps diverging with shrinking size are outside the theorem. Their behavior is not a counterexample to this class. The theorem also does not impose a fixed positive extinction threshold, find an optimal prefactor, or solve unrestricted OQ7.

Mathematical disposition. With an explicit measurable susceptibility hypothesis, the three claims have complete derivations in this audit and no identified invalidating gap. Actual finite spheres attain quartic order with a rigorously nonzero coefficient and controlled remainder. The literal material definition is incomplete, and the prior-art discussion understates close existing formulas. Mathematical validity, numerical corroboration, and scientific novelty are separate conclusions.

A Frozen target and reproducibility record

The original source has 50,946 bytes and SHA-256

`0f4f6167725b10c8fec5479e9dfc562989a4889f13f2d9df2fd3846b422b4572`

The original ZIP has 11,487,001 bytes and SHA-256

`0e46afff09071d367c1702fd3a79f2c0c8f3cf8d7252094da6c593c1b0ec8e42`

The new source archive contains the frozen original ZIP, target source/reference tables, this report's LaTeX/PDF, the new independent Maxwell implementation and controls, their data and logs, the 86-check algebra implementation/results, fresh analytical reviews, source locators, and a hash manifest. Full third-party papers are not redistributed.

After extracting the audit archive, run from its root:

```
python reproduce_audit.py --verify-only
python reproduce_audit.py --full
```

The first checks provenance and saved evidence. The second replays the frozen original pipeline in a temporary extraction, then the independent algebra, Maxwell sweep, and controls. It preserves new run logs and data without modifying the package. Dependencies and commands are described in `README.txt` and `requirements.txt`. The scientific components were executed separately in this audit. The new

wrapper passed syntax, extraction/path preflight, and an actual verification-only invocation; its assembled four-stage replay was not rerun end to end merely because the wrapper was added.

The recorded original replay matched all 25 released cases. The validation summary also verifies four frozen files, 42 original internal checksums, and 16 evidence hashes. A small linear residual or agreement between implementations is not a uniform numerical error bound. No theorem rests on those numerical diagnostics.

The released manuscript remains unchanged. No third party was contacted, and no scientific publication or priority claim was submitted.

References

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[3] B. Hopkins, A. E. Miroshnichenko, A. N. Poddubny, and Y. S. Kivshar, *Fano Resonance Enhanced Nonreciprocal Absorption and Scattering of Light*, Photonics **2** (2015), 745–757. doi:10.3390/photonics2020745. Publisher-provided full text inspected through ResearchGate; Eqs. (12)–(15), Fig. 3.

[4] V. A. Markel, *Extinction, scattering and absorption of electromagnetic waves in the coupled-dipole approximation*, J. Quant. Spectrosc. Radiat. Transfer **236** (2019), 106611. Author accepted manuscript; Eqs. (8), (42), (44)–(49).

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[6] L. V. Brown, H. Sobhani, J. B. Lassiter, P. Nordlander, and N. J. Halas, *Heterodimers: Plasmonic Properties of Mismatched Nanoparticle Pairs*, ACS Nano **4** (2010), 819–832. doi:10.1021/nn9017312. Primary article text inspected on an identified third-party mirror.

[7] Y.-L. Xu, *Electromagnetic scattering by an aggregate of spheres*, Appl. Opt. **34** (1995), 4573–4588. doi:10.1364/AO.34.004573. Publisher abstract/metadata inspected as a method antecedent.

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